Proposal Guidelines: ACL-Style Research Proposal (≈2 pages)

The objective of this proposal is to outline a research project suitable for an ACL paper, grounded in large language models (LLMs) and sequential language data. Your proposal should present a clear research question, justify dataset selection, describe an LLM-based methodology, and articulate a novel contribution. Please note that this is a research proposal, not an implementation report.

1. Project Title, Group name, and group members

Provide a clear, concise ACL-style title that reflects (i) the task or research problem, (ii) the LLM or methodological focus, and (iii) the key contribution.
Example: "Evaluating Long-Context LLMs for Multi-Turn Dialogue in Health Domain Application"

2. Introduction: *(Problem Statement and Motivation)*

Introduce the research problem and clearly define the task you aim to study using LLMs. The task must involve natural language as sequential data, such as document-level text, dialogue, or sequence-to-sequence transformation. Explain why this problem matters from both a scientific and practical perspective. Strong introductions highlight gaps in current LLM research, such as limitations in generalization, robustness, interpretability, data efficiency, or evaluation practices. End this subsection by clearly stating your research question or hypothesis.

3. Related Work

Situate your work within prior ACL main-track research. This section should briefly summarize 4–5 papers and explain how your proposal builds on or differs from them.
Example references (ACL / EMNLP/NAACL main):

- Devlin et al., 2019 introduced BERT, demonstrating the effectiveness of bidirectional pretraining for language understanding tasks and setting a foundation for fine-tuning-based NLP research [1].
- Lewis et al., 2020 proposed BART, a denoising sequence-to-sequence pretraining framework that unifies generation and understanding tasks [2].
- Raffel et al., 2020 introduced T5, framing all NLP tasks in a unified text-to-text format, influencing task formulation for LLMs [3].

Conclude this section by stating what remains unaddressed and how your proposed work fills that gap.

4. Dataset Description and Selection

Describe the dataset(s) you plan to use or construct. You should include:

- Dataset source and domain
- Data format (single-turn text, multi-turn dialogue, long documents)
- Dataset size and label structure (also discuss if you will only use a sample of the data)
- Why this dataset is appropriate for your research question
- Define the **ground truth** (labels/targets) and how it was created (human annotation, metadata, heuristics)

If relevant, discuss preprocessing steps (tokenization, filtering, sequence chunking) and known limitations such as noise, bias, or imbalance.

5. Model and Methodology

Describe the LLM-based approach you plan to use. This may include fine-tuning, prompt-based modeling, instruction tuning, or comparative evaluation across models.
Model details should include:

- Target model(s) (e.g., LLaMA, T5, GPT-style models, open-source transformers)
- Small models to save GPU and memory.
- How the model handles sequential structure (context length, positional encoding, chunking)

Compute and Resource Requirements You must explicitly state:

- Expected GPU type (e.g., Colab, A100)
- Number of GPUs and estimated training or inference time (test a sample).
- Memory requirements (VRAM)
- Whether resources are assumed to be available locally or via cloud platforms

This section demonstrates feasibility and realism, which is critical for ACL-style research.

6. Evaluation Plan

Describe how you will evaluate your model and validate your claims. This may include:

- Quantitative metrics (e.g., accuracy, F1, BLEU, ROUGE)
- Baseline comparisons
- Error analysis or qualitative analysis
- Robustness or generalization experiments

Explain what outcomes would support or refute your hypothesis.

7. Expected Outputs and Novel Contribution

a) Clearly articulate what is new in your work. Novelty may come from:

- A new task formulation or research question
- A new evaluation framework or metric
- A systematic analysis of LLM behavior (e.g., robustness, bias, failure modes)
- A methodological comparison not previously explored

b) List 2–3 concrete outputs, such as:

1. A trained or adapted LLM model with empirical evaluation
2. A curated benchmark or evaluation framework
3. An analysis paper (compare models, prompts, bias) suitable for submission to an ACL venue

8. References: ACL-style citations and include a references section at the end of the proposal.

[1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.* NAACL.
[2] Mike Lewis et al. 2020. *BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension.* ACL.
[3] Colin Raffel et al. 2020. *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer.* ACL.